

Investigation of Fluid Flow Through Capillaries

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This paper investigates the relationship between pressure drop and volumetric flow rate in horizontal capillaries, with the goal of determining when a simple linear model can accurately describe the flow. Using dimensional analysis of six physical variables—fluid viscosity, density, capillary dimensions, pressure drop, and flow rate—a general dimensionless expression was obtained. Assuming slow, steady flow, the drag force was taken to be proportional to fluid velocity, which predicts a linear relationship between pressure drop and flow rate: $\Delta P \propto Q$. This leads to the simplified model $\Delta P = C_p Q \frac{\rho \nu l}{d^4}$.

To test this model, six capillaries with different length-to-diameter ratios were used, and flow rates were measured across a range of pressures created by varying fluid heights. Uncertainties in volume, timing, and pressure were carefully accounted for, and a preliminary test confirmed that any pressure loss in the connecting tube was negligible.

The results confirmed the expected linear relationship between pressure drop and flow rate in the laminar flow regime. The value of C_p was measured to be 41 ± 0.6 for my best fit ($l/d \approx 296$). However, at higher pressures or in wider capillaries, deviations from linearity were observed, resulting in higher errors. This marks the beginning of a second flow regime where the original assumptions no longer apply. In this regime, a simple interpolation of the data suggested a different scaling law, closer to $\sqrt{\Delta P} \propto Q$. While this fit appears to match the curvature of the data, more measurements are needed to confirm the validity of this model.

Overall, the experiment supports the use of the linear drag model for laminar capillary flow and shows how capillary geometry affects resistance. The observation of two flow regimes also highlights the limits of the linear model and the conditions where more complex fluid behavior must be considered.

I. INTRODUCTION

Before conducting any experiments, I took some time to think about the physics at play and tried to answer the question, "what parameters affect the volume flow rate?" in the most general way possible.

I made an important assumption to begin with:

1. My experiments should be conducted in a quasi-stationary state.

A fully stationary state is one where the flow rate through the capillary doesn't change with time, meaning the forces acting in the direction of flow are balanced. By placing my capillary in the horizontal plane, I ensured that the only two forces acting along this direction would be a force due to the pressure drop and a resistive force due to the walls of the capillary. For the following discussion, I'll refer to "the force caused by the pressure drop along the capillary" simply as "the pressure drop" for brevity.

Let's consider for a second what happens if the state is *not* stationary. That is, if the pressure drop were much larger than the resistive force, the water would accelerate along the capillary. Here, I make another assumption:

2. The resistive (drag) force is proportional (in some way or other) to a power of the velocity of water through the tube: $F_d \propto v^\alpha$.

This assumption is consistent with the behavior of drag forces in fluids (like air) seen in classical mechanics, so it seems like a decent starting point.

Now, in our non-stationary state, as the velocity increases, the resistive force also increases, until at some point these two forces equal one another. This state, in which the forces balance ($F_{\text{pressure}} \approx F_{\text{drag}}$), is what I call stationary. This is the regime where I'll be conducting my experiments; of course, the driving pressure decreases slightly during an experiment as the basin empties, hence the term "quasi"-stationary.

The next step was to analyze the capillary itself and hypothesize what parameters could be related to the flow rate. I concluded that for a simple capillary, the only things affecting its flow rate are:

- The properties of the liquid through it: density (ρ), surface tension (γ), and kinematic viscosity (ν).
- The characteristic lengths of the tube: Length (l) and Diameter (d).
- The pressure drop along the tube (ΔP).

These parameters determine the resulting bulk flow rate (Q), which is the volume of fluid passing through the tube per unit time. While Q is what is directly measured, it might be more intuitive to consider the fluid's motion in terms of its average velocity, v . These two quantities are related through the cross-sectional area of the capillary,

$$A = \pi(d/2)^2:$$

$$v = \frac{Q}{A} = \frac{4Q}{\pi d^2} \quad (1)$$

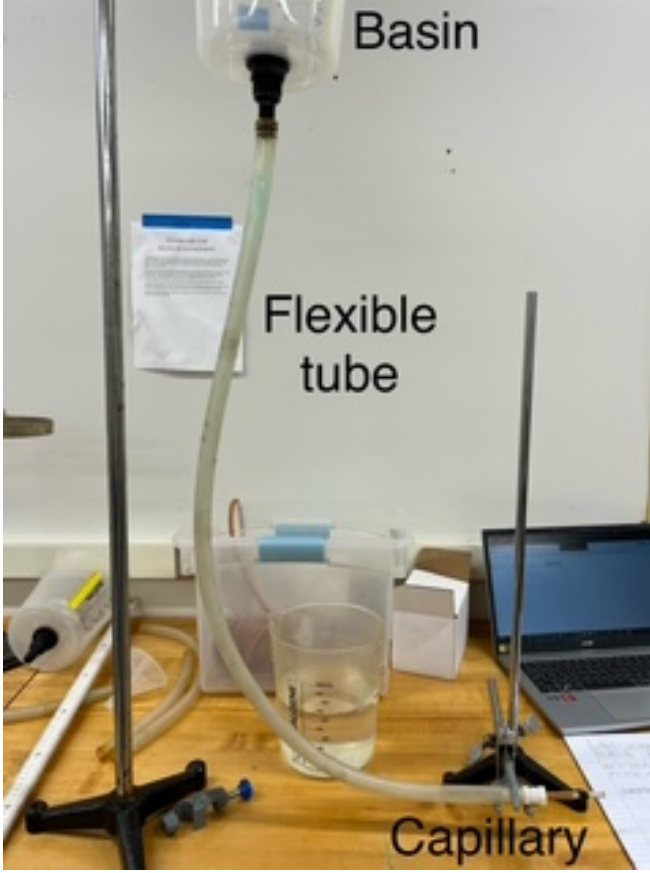


FIG. 1. Experimental setup showing horizontal capillary configuration.

For the dimensional analysis that follows, we will use the bulk flow rate Q .

I assumed that surface tension, which is an "edge" effect, could be neglected. I will discuss this in depth in the Results section below.

I focused on the following six parameters, trying to combine them into one cohesive physical equation:

- Kinematic Viscosity (ν , [m^2/s]) and Density (ρ , [kg/m^3])
- Length (l , [m]) and Diameter (d , [m])
- Pressure (ΔP , [Pa]) and Flow rate (Q , [m^3/s])

My goal was to write the most general dimensionless equation relating these six parameters, as any physical equation can be written in dimensionless form. Using Buckingham's Pi theorem [6], which states that the number of independent dimensionless groups equals the number of parameters minus the number of fundamental dimensions (in my case, six parameters and three dimensions: Mass, Length, Time), the most general relation

between these six quantities can be written as:

$$\frac{\Delta P d^4}{\rho Q^2} = f\left(\frac{l}{d}, \frac{\nu l}{Q}\right) \quad (2)$$

Which can be rewritten explicitly for pressure:

$$\Delta P = \frac{\rho Q^2}{d^4} f\left(\frac{l}{d}, \frac{\nu l}{Q}\right) \quad (3)$$

Now here is where we make our final crucial assumption, supplementing Assumption 2:

3. Under my experimental conditions, the drag force is *linearly* proportional to the average fluid velocity ($F_{\text{drag}} \propto v$).

This assumption was motivated by the analogy to drag in mechanics, where we learned that for sufficiently low velocity of a body, the drag force is linearly proportional to the velocity, $F_{\text{drag}} \propto v$. When an object reaches terminal velocity, $F_{\text{grav}} = F_{\text{drag}} \propto v$.

In our setup, the force caused by the pressure drop is balanced by the drag force. From our quasi-stationary assumption, $F_{\text{pressure}} = F_{\text{drag}}$. Since $F_{\text{pressure}} \propto \Delta P$ and we are now assuming $F_{\text{drag}} \propto v$, and knowing $v \propto Q$ (from Eq. (1)), it follows directly that:

$$\Delta P \propto Q \quad (4)$$

This assumption of linearity greatly simplifies our general Eq. (3). If $\Delta P \propto Q$, then the function f must contain a factor that cancels one power of Q . Since f depends on Q only through the second argument, $\nu l/Q$, the function must be directly proportional to this term. We can therefore write $f(l/d, \nu l/Q) = h(l/d) \cdot (\nu l/Q)$, where $h(l/d)$ is a new unknown function depending only on the aspect ratio. Substituting this into Eq. (3) and simplifying gives our working model:

$$\Delta P = Q \frac{\rho \nu l}{d^4} h\left(\frac{l}{d}\right) \quad (5)$$

The primary goal of the following experiments is to measure the value of this function, $h(l/d)$, and to determine the conditions under which this simple linear model holds true. To do this I chose six ratios (see Table I).

TABLE I. Capillary dimensions used in experiments.

Trial	Length (cm)	Diameter (cm)	Ratio (l/d)
1	5.05	0.19380	26.1
2	5.1	0.1016	50.2
3	10.0	0.1016	98.4
4	15.25	0.07366	207.0
5	15.2	0.05131	296.2
6	15.2	0.03048	498.7

Now before any experiments could be conducted, I had to make another important assumption. The assumption

was that the pressure drop along the capillary was equal to the pressure drop caused by the height of the water above the opening of the capillary:

$$\Delta P = \rho gh \quad (6)$$

This would be true with a sufficiently short and wide flexible tube and short and wide nozzles, where the pressure drop among these items could be neglected. The results of this experiment can be found in the Methods section below.

II. METHODS

In this experiment, measurements came down to measuring how much water (in cm^3) came out of the capillary in some amount of time t (seconds) and dividing the two. Each flow rate, q , would then correspond to a pressure, ΔP , which I measured by finding the height h above the end of my capillary and the highest level of the water in my water basin. So there were three main sources of error: Timing, Volume, and Height. Physical constants such as the density [1], viscosity [2], surface tension [3] of water, and the acceleration due to gravity [4] were taken from established sources.

As a general rule of thumb, I ensured that I would "purge" the flexible tube before any experiment, so as to minimize the error caused by bubbles present at the opening of the nozzles connecting the tube to the capillary or basin. I made sure to also let the capillary run for a bit before starting to collect data, so as to ensure my quasi-stationary state was satisfied.

As for timing error, I tried to time each run for at least 30 seconds; my longest one measured just over 45 minutes for 60 ml, and my shortest one measuring just over 10 seconds for 80 ml. I measured my reaction time on humanbenchmark.com/tests/reactiontime [5], and found it to be about 0.25 seconds. Seeing as this test sees how well you can react to external factors, my timing error was probably smaller than this, as I got to decide when I'd start timing (by placing my water collecting receptacle below the end of the capillary) and when I stopped timing (quickly moving the water receptacle out from under the stream and stopping the timer simultaneously). I therefore adopted a conservative uncertainty of $\delta t = 0.25$ s for all trials.

I measured the volume of water in two ways.

When the flow rate was large, measuring the mass of the water that came out and dividing by the very well-known density of water made sense. For measurements like these, I measured about 80 ml of water per trial (corresponding to about 80 g of water collected). My scale was good up to 1 g, thus my volume uncertainty for these trials was $\delta V = 1.0 \text{ cm}^3$. For most trials run in this way, I collected a few measurements per pressure and averaged them in order to reduce random errors.

For trials where the flow rate was small, it would theoretically take hours to collect 80 ml, so I opted for a

simpler approach. I measured about 10 ml of water using a precise volumetric flask, with an uncertainty of about $\delta V = 0.2 \text{ cm}^3$. This greatly increased my ability to measure flow rate for the very thin diameters, but still took a considerable amount of time.

The uncertainty in $q = V/t$ therefore depends on my uncertainties in measuring volume (δV) and time (δt). The formal way to combine these is with partial derivatives [7]:

$$\delta q = \sqrt{\left(\frac{\partial q}{\partial V} \delta V\right)^2 + \left(\frac{\partial q}{\partial t} \delta t\right)^2}$$

The partial derivatives of the flow rate $q(V, t)$ are $\frac{\partial q}{\partial V} = \frac{1}{t}$ and $\frac{\partial q}{\partial t} = -\frac{V}{t^2}$. Substituting these into the formula gives the instrumental uncertainty, δq_{inst} , for any single trial based on its measured volume V and time t :

$$\delta q_{\text{inst}} = \sqrt{\left(\frac{\delta V}{t}\right)^2 + \left(\frac{V \delta t}{t^2}\right)^2} \quad (7)$$

For pressure settings where I took N multiple trials, the plotted flow rate is the mean value, $\bar{q}$. The uncertainty in this mean, $\delta \bar{q}$, has two components. The first is the statistical uncertainty (the standard error of the mean), $\delta q_{\text{stat}} = s_q / \sqrt{N}$, where s_q is the standard deviation of the N flow rate measurements. The second is the instrumental uncertainty, which I calculated using Eq. (7) with the average volume and time from all experiments. The final error for the plotted point is the quadrature sum of these two sources: $\delta \bar{q} = \sqrt{(\delta q_{\text{stat}})^2 + (\delta q_{\text{inst}})^2}$.

Similarly, the error for my linear velocity v is found from the error in the mean flow rate $\delta \bar{q}$. Since $v = \bar{q}/A$ and the cross-sectional area $A = \pi d^2/4$ is a constant (assuming negligible uncertainty in d):

$$\delta v = \left| \frac{\partial v}{\partial \bar{q}} \delta \bar{q} \right| = \frac{1}{A} \delta \bar{q} = \left(\frac{4}{\pi d^2} \right) \delta \bar{q} \quad (8)$$

For my height errors (which correspond to my pressure errors), I was good up to the millimeter of the measuring stick. Since I measured first the height from the table to the end of the capillary, and again from the table to the top of the water, my total measurement uncertainty from the ruler was $\delta h_{\text{ruler}} = 0.2 \text{ cm}$.

Now, in my experiments I did not assume that the height of the water in the basin dropped negligibly. About 88 ml of water collected corresponded to a drop of about 0.5 cm in the height of the basin. A much lower error here would have been achieved with a much wider basin (ours was only about 15 cm in diameter). For pressures that were caused by a height difference on the order of magnitude of 50 cm, a 0.5 cm drop in height could be neglected, but for a difference of 5 cm in height, this drop is significant. This is why for measurements like these, the volumetric flask served as a great tool to minimize this effect.

Thus, the pressure uncertainty has two sources:

1. The measurement uncertainty from the ruler, $\delta h_{\text{ruler}} = 0.2$ cm.
2. The uncertainty from the drop of the water level, Δh .

Based on my calibrations: for 88 ml collected, $\Delta h \approx 0.5$ cm; and for ~ 9 ml collected, $\Delta h \approx 0.05$ cm. This drop introduces a systematic effect, which I correct for by using the average height $h_{\text{avg}} = h_{\text{initial}} - \Delta h/2$. The uncertainty in this correction is $\delta h_{\text{drop}} = \Delta h/2$.

These two uncertainties are combined in quadrature to find the total uncertainty in h :

$$\delta h_{\text{total}} = \sqrt{(\delta h_{\text{ruler}})^2 + (\delta h_{\text{drop}})^2} \quad (9)$$

Since the pressure is calculated as $\Delta P = \rho gh$, where we take ρ and g as having negligible error relative to h , the uncertainty in pressure is found by propagating δh_{total} [7]:

$$\delta P = \left| \frac{\partial P}{\partial h} \delta h_{\text{total}} \right| = \rho g \delta h_{\text{total}} = \rho g \sqrt{(\delta h_{\text{ruler}})^2 + (\delta h_{\text{drop}})^2} \quad (10)$$

This method correctly accounts for the fact that my low-height measurements, which also used small volumes, have a smaller error from the height drop, while all measurements retain the 0.2 cm ruler uncertainty.

Preliminary Test: Neglecting the Pressure Drop in the Flexible Connecting Tube

A potential source of error in assuming $\Delta P = \rho gh$ is the pressure drop across the flexible connecting tube itself. If this drop were not negligible compared to the pressure drop across the capillary, it would introduce a systematic error dependent on flow rate and tube length. I therefore conducted a preliminary experiment to test this assumption.

To do this, I set up a "worst-case scenario." The tube's own pressure drop should be most noticeable when the capillary's resistance is as small as possible. I therefore picked the capillary with the lowest resistance I have: the widest one (diameter $d = 0.1938$ cm) and the shortest one ($l = 5.1$ cm). My thinking was that if the tube's effect was negligible here, it would certainly be negligible when I switched to the narrower, longer capillaries, whose own resistance would dominate the system.

I measured the flow rate for three different tube lengths, keeping the height of the apparatus constant at about 41 cm. The results are shown in Table II. (It is worth noting that a trial with a 49.5 cm tube was discarded as a procedural error during that run rendered the data unreliable).

To determine if the differences in the means are statistically significant, I must compare them to the measurement uncertainty. For these specific runs, a typical measurement was $V \approx 100$ cm³ in $t \approx 20$ s. Using the

TABLE II. Measured flow rates for connecting tubes of varying lengths.

Length (cm)	Flow Rate Trials (cm ³ /s)					Mean $\bar{Q}$ (cm ³ /s)
	1	2	3	4	5	
31.5	4.812	4.818	4.802	4.821	4.822	4.815
65.0	4.828	4.832	4.803	4.804	4.844	4.822
91.0	4.778	4.816	4.816	4.756	4.762	4.786

instrumental error from Eq. (7) with $\delta V = 1.0$ cm³ and $\delta t = 0.250$ s:

$$\delta q_{\text{inst}} = \sqrt{\frac{(1)^2}{(20)^2} + \frac{(100)^2(0.250)^2}{(20)^4}}$$

$$\delta q_{\text{inst}} = \sqrt{0.0025 + 0.00390625} \approx 0.080 \text{ cm}^3/\text{s}$$

Thus, the uncertainty for any single measurement in this test is about ± 0.080 cm³/s. The most direct test is to compare the difference between the extreme cases (the shortest and longest tubes) with this uncertainty. The measured effect is:

$$\Delta q_{\text{effect}} = |q_{31.5\text{cm}} - q_{91\text{cm}}| = |4.815 - 4.786| = 0.029 \text{ cm}^3/\text{s}$$

This measured effect of 0.029 cm³/s is less than half the magnitude of the experimental uncertainty (± 0.080 cm³/s). Based on this, I can confidently say that the connecting tube's resistance is negligible compared to the capillary resistance for the range of parameters studied, and I don't need to worry about its length in my main experiments.

III. RESULTS

After developing a baseline level of theory for my experiment, I thought it wouldn't be too difficult to get into my "linear" regime. The whole goal of my experiment was to test capillaries and vary pressures until graphing pressure ΔP vs flow rate Q made a linear plot, corresponding to a log-log power slope of 1. Turns out I was wrong – it isn't easy to get linearity.

As seen in Table I, my largest ratio tested was ≈ 500 and the smallest was ≈ 26 . These ratios were the corresponding maximum and minimum ratios I had access to, and I reasoned I would test a range of ratios between these two so that I could model $h(l/d)$ with at least 5 different aspect ratios.

Turns out it was difficult to get into the linear regime, and my desire to have 5 'good' points was ambitious in the time I had to conduct tests. After the preliminary flexible tube tests, I decided that I would continue testing my shortest and widest capillary (Capillary 1 in Table I, $l/d \approx 26$). As I tested, I continued plotting points as I went along, and I clearly noticed non-linearity in the Q vs ΔP plot:

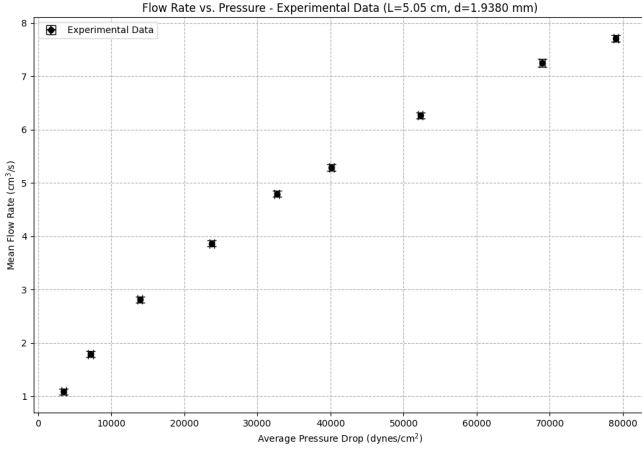


FIG. 2. Scatter plot of Q vs ΔP for Capillary 1 ($l/d \approx 26$).

There will be further discussion on what can be analyzed from this graph, and it leads to some interesting conclusions, but for now let's continue with the original plan: achieve linearity and find $h(l/d)$.

Going down in diameter and increasing length, I was able to achieve a wide range of ratios. I ended up being able to use four ratios of l/d that resulted in decently linear plots, which I could then use to calculate my h value. Remember from our theoretical development, combining the dimensional analysis (Eq. (3)) with the assumption of linearity ($\Delta P \propto Q$) led us directly to Eq. (5):

$$\Delta P = Q \frac{\rho \nu l}{d^4} h \left(\frac{l}{d} \right)$$

This theoretical model predicts a linear relationship between the pressure drop ΔP and the flow rate Q , passing through the origin. Experimentally, we plot Q versus ΔP and determine the slope, $m = dQ/d(\Delta P)$. According to our model in Eq. (5), this slope m is related to $h(l/d)$ as follows:

$$\frac{d(\Delta P)}{dQ} = \frac{1}{m} = \frac{\rho \nu l}{d^4} h \left(\frac{l}{d} \right) \quad (11)$$

Therefore, we can calculate our experimental value for $h(l/d)$ for each capillary by rearranging this equation and using the experimentally measured slope m from the Q vs P graph:

$$h \left(\frac{l}{d} \right) = \frac{d^4}{m \rho \nu l} \quad (12)$$

The four plots that got me the best linearity are shown below (Figure 3), with the corresponding l/d ratio, the slope m of the Q vs P line, and the R^2 value. The equation above was used to determine the experimental values of $h(l/d)$ from the slopes obtained in the linear fits shown below.

Now is perhaps a good time to talk about how the error for the slope was calculated. The linear fits were performed using the `scipy.optimize.curve_fit` function

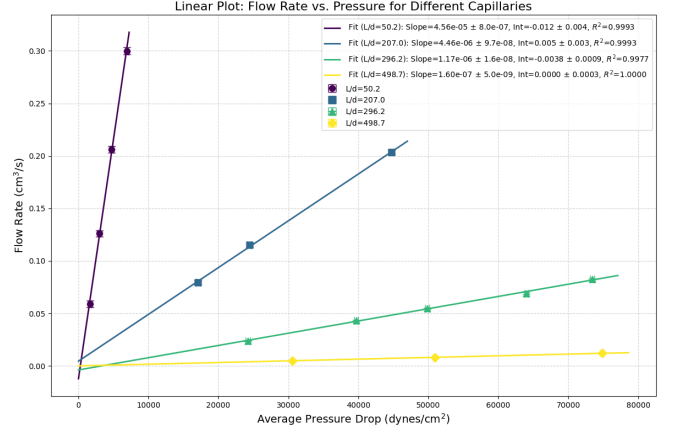


FIG. 3. Linear fits (Q vs ΔP) for the four capillaries exhibiting approximately linear behavior.

[10] in Python [13]. This algorithm performs a weighted least-squares fit, taking into account the uncertainties in both flow rate Q (propagated from volume and time errors as discussed in the Methods section) and pressure ΔP . The output of `curve_fit` includes a covariance matrix, the square root of whose diagonal elements provides the standard error (uncertainty) in the fitted parameters, namely the slope m and the intercept b . This uncertainty reflects how much the slope could vary while still being consistent with the data and its associated error bars.

After calculating the slope m and its uncertainty δm in this way, I used each slope from the linear plots to find $h(l/d)$ by using Eq. (12). The uncertainty in h was propagated from the uncertainty in the slope δm [7].

Plotting the four calculated values of $h(l/d)$ we see the following:

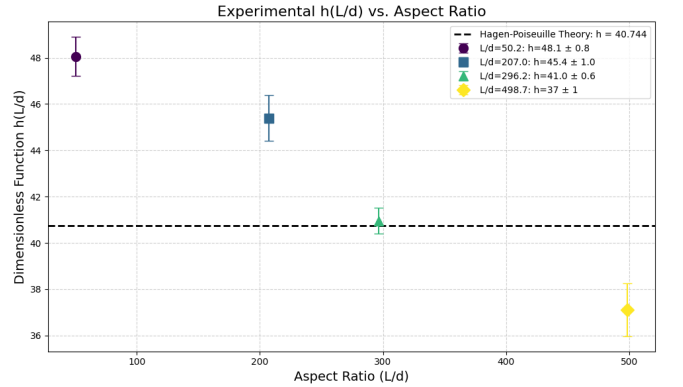


FIG. 4. Experimental values of $h(l/d)$ vs. aspect ratio l/d .

At this point, it's useful to revisit the theory. Although the following argument logically belongs in the introduction with the dimensional analysis, I reached this conclusion only after analyzing the experimental $h(l/d)$ values. Presenting it here reflects the actual progression of the analysis. This final physical argument helps clarify the

experimental results and ultimately determines the expected form of the function $h(l/d)$. To remind you of some of the basic assumptions in this experiment: I assumed that I was working in a quasi-stationary regime, where the only two forces along the capillary were the one caused by the pressure drop and that of the resistive drag force. The drag is a consequence of the viscous shear between the fluid and the inner walls of the capillary. It is therefore reasonable to assume that the total drag force must be proportional to the inner surface area of the tube, which is πdl . Thus, the drag force must be linearly proportional to the length of the capillary, $F_{\text{drag}} \propto l$.

Since the pressure force must balance this drag in our quasi-stationary state ($F_{\text{pressure}} = F_{\text{drag}}$), it follows that the pressure drop must also be linearly proportional to the length of the capillary:

$$\Delta P \propto l \quad (13)$$

Let us now re-examine our derived model from Eq. (5):

$$\Delta P = \left(\frac{\rho \nu Q}{d^4} \right) \cdot l \cdot h \left(\frac{l}{d} \right)$$

For this equation to be consistent with our physical requirement that $\Delta P \propto l$, the function $h(l/d)$ cannot introduce any additional dependence on l or d . The only way for the model to remain consistent is if the function $h(l/d)$ is, in fact, not a function of the aspect ratio at all, but rather a dimensionless constant.

This is a major step.

Let us call this new dimensionless constant C_p . This leads to our final, most specific theoretical model for laminar flow through a capillary:

$$\Delta P = C_p \frac{\rho \nu l}{d^4} Q \quad (14)$$

Looking back at the figure above showing $h(l/d)$ vs l/d (Figure 4), it is probably clear that this plot doesn't exactly resemble what theory makes us hope for (a constant value). But what exactly *is* the correct theory?

The relation that I derived in Eq. (14) is Poiseuille's Law [11], which models the "laminar capillary flow regime" – the same regime that I was looking for in which ΔP was linearly proportional to Q . The standard equation for Poiseuille's Law, expressed in terms of kinematic viscosity ($\mu = \rho \nu$) and diameter ($R = d/2$), is:

$$\Delta P = \frac{8\mu l Q}{\pi R^4} = \frac{8(\rho \nu) l Q}{\pi (d/2)^4} = \frac{128}{\pi} \frac{\rho \nu l}{d^4} Q \quad (15)$$

Comparing Eq. (15) with my equation from Eq. (14), we see that the theoretical value of the constant C_p should be $128/\pi$, which is about 40.74.

In the $h(l/d)$ plot from before (Figure 4), we notice a few things. Firstly, we notice that my $h(l/d)$ plot doesn't exactly have a slope of 0, and actually looks like it trends downward. This can be explained. In the same sense that

I found it difficult to get into the linear regime with larger diameters (lower l/d), it was difficult to get a "correct" value for h when the diameter was larger. i.e., the less laminar my regime was (and therefore the less linear the Q vs P plot), the more my calculated slope $m = dQ/dP$, and thus my calculated $h \propto 1/m$, deviated from the theoretical Poiseuille value.

It could be expected that the linear regime is achieved as l/d increases and the flow rate decreases. In that case, Eq. (14) is achieved asymptotically as $l/d \rightarrow \infty$. Meaning that as I increased the l/d ratio and got closer and closer to my truly laminar regime, I started getting closer to the true value of $h \approx 40.74$. The green point, with an aspect ratio of ($l/d \approx 296$) seems to fit to the theoretical value the best. But how does this explain the point at the highest ratio of l/d ? After all, we'd expect this point to resemble the theory the best, as it had the most "laminar" flow, but it clearly doesn't. In fact this point, with a diameter of 0.03048 cm ($l/d \approx 499$), has an h value of ≈ 37 , less than the theoretical value. Let's look at a plot of the linear fit for this diameter:

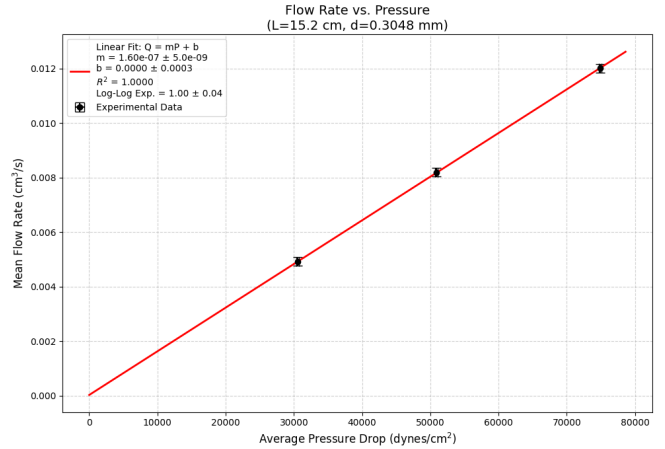


FIG. 5. Linear plot of Q vs ΔP for the smallest diameter capillary ($d = 0.03048$ cm).

This plot is virtually as close to perfect as I could hope for: an R^2 value near 1, a near-zero intercept, a log-log exponent close to 1, and data which falls almost exactly on the line. Granted, three data points isn't necessarily "enough" for a good linear plot, but the plot appears almost ideal besides this. To explain why I believe this capillary gave underestimates of our C_p value, it is necessary to reexamine some assumptions I made about surface tension and gravity, which I promised to circle back to.

Edge Effects: Negligibility of Surface Tension and Gravity

There are two dimensionless groups, constructed using similar approaches to those discussed in the introduction,

that can help analyze edge effects related to gravity and surface tension (γ).

The first group compares the effect of gravity pulling down on the water to the effect of surface tension holding it together: $\pi_1 = \rho g d^2 / \gamma$. If π_1 is much larger than 1, it suggests gravity is strong enough to significantly deform the water's surface shape compared to surface tension. For the wider tubes, where I didn't expect surface tension to be as dominant, this dimensionless group is around unity, indicating that both gravity and surface tension play comparable roles for the water's shape at the exit of the widest tube. For example, using the widest tube ($d = 0.19380$ cm):

$$\pi_1 = \frac{\rho g d^2}{\gamma} = \frac{(0.9978 \frac{\text{g}}{\text{cm}^3})(980.7 \frac{\text{cm}}{\text{s}^2})(0.19380 \text{ cm})^2}{72.4 \frac{\text{dyn}}{\text{cm}}} \approx 0.51$$

The second group compares the tendency of the out-flowing water to shoot out straight from the capillary due to its speed, to the tendency of gravity to pull it downwards: $\pi_2 = Q^2 / (g d^5)$. For gravity's pull at the tube's exit to be negligible, we need $\pi_2 \gg 1$, or $Q \gg \sqrt{g d^5}$. In cases like the wider tubes, where π_1 was near unity (meaning gravity was relevant for the shape), it was necessary that the flow rate be sufficiently large so that the water's motion dominated gravity at the outlet, ensuring the tube remained effectively full and didn't just dribble out. This was generally true for wider tubes.

For thinner capillaries, the first dimensionless group becomes much smaller than unity, indicating that surface tension strongly dominates gravity in maintaining the cylindrical water shape and ensuring the tube is full. Since $\pi_1 \ll 1$, surface tension guarantees the tube remains full. It wasn't strictly necessary that $Q \gg \sqrt{g d^5}$, although it was mostly true.

Now here is where the outlier from the plot of $h(l/d)$ can be explained. While the large surface tension in the thinner capillaries helped ensure the tube was full, it doesn't necessarily mean surface tension doesn't affect the flow rate in other ways. For this, we have to look at the Laplace pressure [12]. The Laplace pressure, P_L , is the pressure difference across the curved surface of a liquid due to surface tension. It is given by $P_L = 4\gamma/d$ for a cylindrical tube of diameter d . This pressure can either oppose or aid the flow – effectively acting like an additional pressure drop or pressure boost. For the $d \approx 0.3$ mm capillary ($d = 0.03048$ cm), the Laplace pressure is:

$$P_L = \frac{4\gamma}{d} = \frac{4(72.4 \text{ dyn/cm})}{0.03048 \text{ cm}} \approx 9500 \text{ dynes/cm}^2$$

This value is on the same order of magnitude as the pressures from the water height I was measuring flow rates at for this tube (which ranged from roughly 30000 to 73000 dynes/cm²). I suspect that the diameter of ≈ 0.3 mm experienced a significant wetting effect, where the surface tension effectively "pulled" extra water out of the tube, acting like an additional driving pressure. This would increase the flow rate Q for a given pressure ΔP caused by

the basin height. This makes sense why the calculated value of h is smaller than expected, as $h \propto 1/m$ (where m is the slope dQ/dP), and m increases if "more water" than expected gets pulled out.

In short, to get desirable results for this experiment, the diameter needs to be small enough (large l/d) to ensure laminar flow and minimize entrance effects, but not so small that surface tension effects (like Laplace pressure) become dominant and interfere with the purely pressure-driven flow described by Poiseuille's Law.

IV. CONCLUSION

My goal was to correctly characterize the linear flow regime of water through a capillary. In doing so, I came up with a theoretical model based on dimensional analysis and physical arguments, which predicted that for laminar flow:

$$\Delta P = C_p \frac{\rho \nu l}{d^4} Q \quad (16)$$

where C_p is a dimensionless constant expected to be independent of the capillary's geometry. The experiments aimed to measure this constant and verify the conditions under which this linear model holds.

My best capillary, with an aspect ratio of 296.2, which showed good linearity ($R^2 > 0.999$) and which was presumably least affected by edge or entrance effects, gave a C_p value of:

$$C_p = 41.0 \pm 0.6$$

Compared to the theoretical prediction from Poiseuille's Law [11], $C_{p,\text{theory}} = 128/\pi \approx 40.74$, this capillary has a percent deviation of only:

$$\text{Percent Deviation} = \frac{|41.0 - 40.74|}{40.74} \times 100\% \approx 0.6\%$$

This excellent (yet potentially cherry-picked) agreement supports the validity of the Poiseuille model in the appropriate regime.

Of course, the agreement was not always this good across all capillaries tested. My measured C_p values ranged significantly, with the error reaching as high as 18 percent for the smallest ratio ($l/d \approx 50$) and 9 percent for the largest ratio ($l/d \approx 499$). These larger deviations highlight the presence of other physical effects not accounted for in the simple laminar flow model, particularly when operating outside the optimal geometric range.

From Figure 4 There might be a clear indicator of how to vary our capillaries in order to get into this laminar regime the best, so as to be more confident in our value of C_p . As you go down in diameter (increase l/d), it is clear that it is "easier" to get deeper into the laminar regime, as indicated by the experimental h values approaching the theoretical constant C_p . It is also clear from the

individual capillary plots that as your flow rate decreases (lower pressure), you get deeper into this regime too.

But just *how* small the diameter and *how* slow the flow must be to truly reach this linear regime, and whether there are other factors at play that characterize the cut-off from laminar to turbulent flow, are still unanswered questions. I'll return to this after discussing the non-linear regime we saw earlier, as promised.

In revisiting this non-linear regime, let's look back at Figure 2 from before, which used the data from Capillary 1 ($l/d \approx 26$). It was evident that Q vs P wasn't linear, which is why I didn't include this capillary in my analysis for the laminar model constant C_p .

Plotting the points from Figure 2 on a log-log plot, we hope to get some idea of what model might fit the data across different flow speeds. We know that as the diameter decreases (and l/d increases), we get more linear. Let's divide the data for Capillary 1 into two sets: the two points corresponding to the lowest pressures (below ≈ 10000 dynes/cm²), and the other seven points at higher pressures:

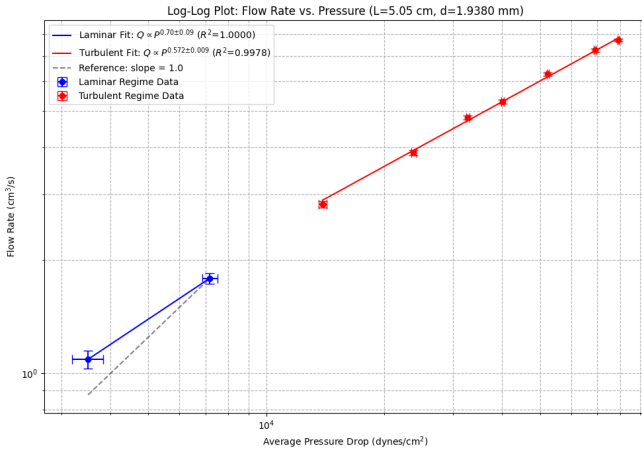


FIG. 6. Logarithmic plot of Q vs ΔP for Capillary 1 ($l/d \approx 26$), split into two regimes.

Fitting yields that the regime on the left (which I'll call quasi-laminar for this tube) has a slope of about $m \approx 0.7$ – clearly not quite linear yet. The regime on the right, however, yields a log-log power law exponent of $m \approx 0.57$. Now this is interesting. Of course, I believe that the regime on the left isn't truly laminar quite yet, so fitting it to a linear plot would be misleading, but we'll do so anyways to compare. The regime on the right shows some promise of a clearly different power law between flow rate and pressure.

I hypothesize that the relationship between flow rate and pressure in the turbulent regime follows something like $Q \propto \sqrt{P}$, or $Q \propto P^{1/2}$. This far-fetched guess would, however, make sense in light of our air drag analogy. We know that at very high velocities, the drag force is often proportional to v^2 instead of v . Assuming that in the turbulent regime the force caused by the pressure drop is

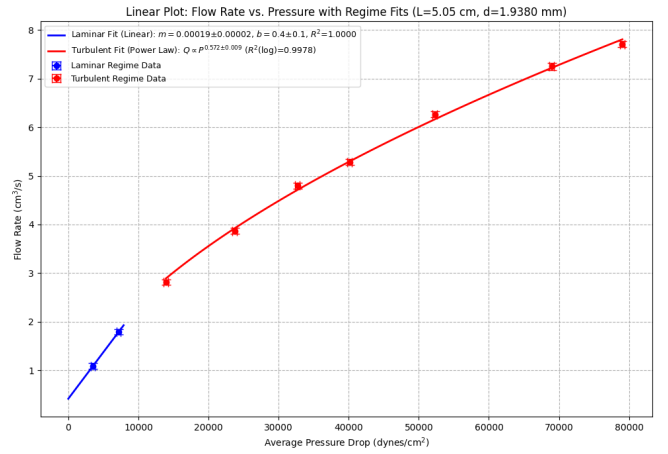


FIG. 7. Linear plot of Q vs ΔP for Capillary 1 ($l/d \approx 26$), with regime fits overlaid.

still balanced by the drag force ($F_{\text{pressure}} = F_{\text{drag}}$), and if $F_{\text{drag}} \propto v^2$, then we can make a guess that the pressure drop is proportional to v^2 . Since $v \propto Q$, this implies $\Delta P \propto Q^2$, or $Q \propto \sqrt{\Delta P} = (\Delta P)^{1/2}$. Our experimental exponent of ≈ 0.57 for the higher flow rates is remarkably close to this $1/2$ prediction. Of course, this is just a hypothesis, as I only have one dataset exhibiting clear non-linearity to quantify this model, but it would be interesting for future experiments to get deeper into this turbulent regime, either by increasing pressure or diameter, to test this $Q \propto \sqrt{P}$ relationship more rigorously.

There is another capillary (Trial 3 in Table I, $l/d \approx 98$) which I haven't yet talked about. The data I collected wasn't quite linear enough to be clearly laminar, and not quite curved enough to be definitively turbulent:

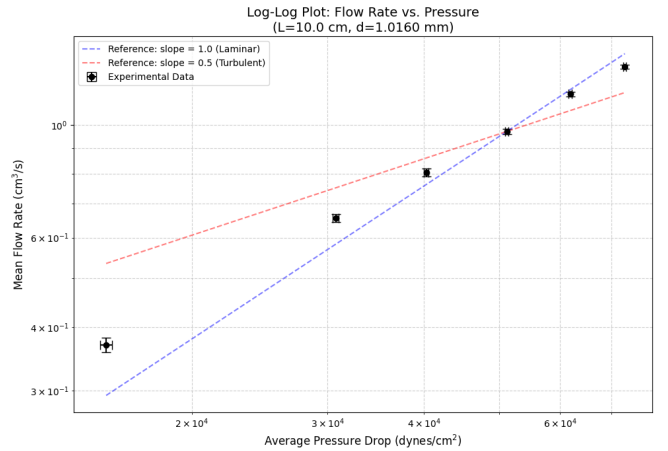


FIG. 8. Log-Log plot of Flow Rate vs. Pressure for the capillary with $l/d \approx 98$, showing behavior intermediate between the laminar and turbulent regimes.

There is perhaps some transitory regime between purely laminar and fully turbulent flow. As hinted at before, I guess that this transition is governed by at least the

speed of the water flow and the diameter of the capillary. That is, more laminar flow probably corresponds with slower flow and smaller diameters, while turbulent flow seems characterized by faster flow through wider capillaries.

So how would I get better values for C_p ? Answering this question will also help us understand the weird transition we see above. From fluid dynamics, we already know that one group that dictates the switch from laminar to turbulent flow is the Reynolds number[14]:

$$Re = \frac{\rho v d}{\mu} = \frac{4\rho Q}{\pi\mu d} \quad (17)$$

Using Poiseuille's Law (Eq. (15)) to substitute for $Q \approx \frac{\pi d^4 \Delta P}{128\rho\nu l}$, we can rewrite the Reynolds number in a more helpful form relating it directly to the pressure drop:

$$Re \approx \frac{4\rho}{\pi\mu d} \left(\frac{\pi d^4 \Delta P}{128\rho\nu l} \right) = \frac{\rho\Delta P d^3}{32\mu^2 l} \quad (18)$$

where $\mu = \rho\nu$ is the dynamic viscosity. Now, while my data doesn't definitively prove that the switch is governed solely by this number, we have already seen from a few perspectives that higher velocity (higher ΔP) and larger diameter cause more turbulent flow, while the opposite is true for laminar. To get better values for C_p , it would be necessary to minimize the Reynolds number ($Re \ll 1$, ideally) while also ensuring that the Laplace pressure effect from before is sufficiently small. Thus, we have a second dimensionless group to consider, let's call it the "pressure ratio":

$$\Pi_P = \frac{P_L}{\Delta P} = \frac{4\gamma}{d\Delta P} \quad (19)$$

To get the best results (i.e., to be in the most "linear," Poiseuille-like regime), we'd need to ensure both of these dimensionless groups are much less than 1 ($Re \ll 1$ and $\Pi_P \ll 1$).

The question becomes: For a given fluid (water, in my case), what can I change in my experiment to get into this linear regime best? Well, the only 'knobs' I can turn in my apparatus are the length (l) and diameter (d) of the capillary and the height (h) of the apparatus (and therefore the pressure ΔP).

Let's look at the trade-offs:

- Decreasing the diameter d : This decreases the Reynolds number ($Re \propto d^3$, good), but increases the pressure ratio Π_P ($\Pi_P \propto 1/d$, bad).

- Decreasing the pressure ΔP (by decreasing height h): This decreases the Reynolds number ($Re \propto \Delta P$, good), but increases the pressure ratio Π_P ($\Pi_P \propto 1/\Delta P$, bad).

So obviously, changing the height knob or the diameter knob arbitrarily isn't the best way to get squarely into our desired linear regime while satisfying both conditions.

However, rearranging the equation for the Reynolds number (Eq. (18)), we see that for a given liquid, capillary diameter, and apparatus height (pressure), $Re \propto 1/l$. Increasing the length of our capillary doesn't affect the pressure ratio Π_P (which only depends on d and ΔP), but it directly decreases our Reynolds number (good!).

Thus, for future experiments aiming for a more precise measurement of C_p , I'd try to get my hands on longer capillaries (I already used the longest ones readily available to me). Longer capillaries should allow operation at sufficiently high pressures (to keep Π_P small) while still maintaining a low Reynolds number, therefore getting into the truly laminar regime better.

It would be interesting for a future experiment to systematically map out this transition by measuring the flow rate for various combinations of l , d , and ΔP , perhaps using different fluids as well. This could help establish a clearer boundary, possibly defined by a critical Reynolds number, marking where the flow behavior shifts away from the simple linear model.

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